

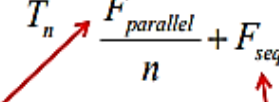
CSS311- Parallel and Distributed Computing (PDC)

Amdahl's law is an expression used to find the maximum expected improvement to an overall system when only part of the system is improved.

- Used in parallel computing to predict the theoretical maximum speedup using multiple processors.

Amdahl's law

- If you put in n processors, you should get n times Speedup (and 100% Speedup Efficiency), right? **Wrong!**
- There are **always some fraction of the total operation** that is **inherently sequential** and **cannot be parallelized** no matter what you do. This includes **reading data, setting up calculations, control logic, storing results, etc.**
- Assume that **all the operations** that **a program** needs to do as **being divided between** a fraction that is **parallelizable** and a fraction that isn't (being sequential), then Amdahl's Law says.

$$Speedup_n = \frac{T_1}{T_n} = \frac{1}{\frac{F_{parallel}}{n} + F_{sequential}} = \frac{1}{\frac{F_{parallel}}{n} + (1 - F_{parallel})}$$


This Fraction Can be reduced by employing multiple processors

This Fraction Can't divided

Amdahl's law

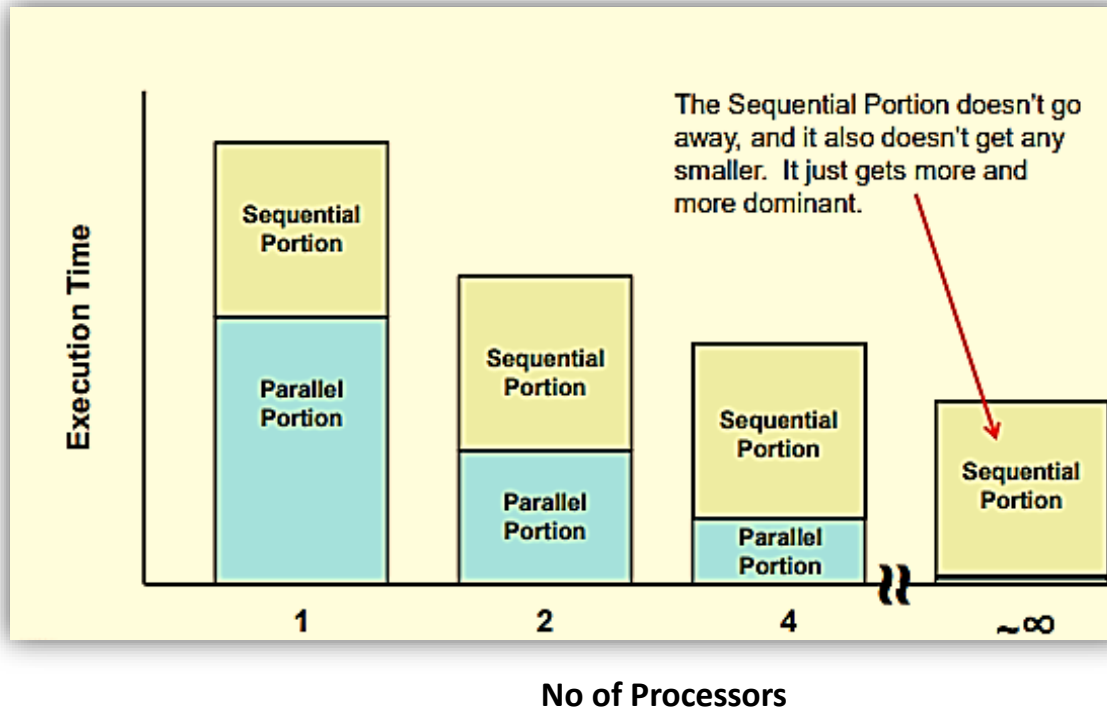
The Table shows the value of speed up for different values N and P.

Table suggests that speed up increases as P increases

N	Speedup		
	P = .50	P = .90	P = .99
10	1.82	5.26	9.17
100	1.98	9.17	50.25
1000	1.99	9.91	90.99
10000	1.99	9.91	99.02

Amdahl's law

Pictorial representation of Amdahl's Law



A program (or algorithm) which can be parallelized can be split up into two parts:

A part which cannot be parallelized

A part which can be parallelized

Amdahl's law

- Amdahl's law is an expression used to find the maximum expected improvement to an overall system when only part of the system is improved.
- It is often used in parallel computing to predict the theoretical maximum speedup using multiple processors.

Expression

$$\text{Speedup} = \frac{\text{Execution Time Before Improvement}}{\text{Execution Time After Improvement}}$$

Amdahl's law is generally written as:

$$\text{Speedup} = \frac{1}{(1 - \text{fraction}_{\text{Enhanced}}) + \left(\frac{\text{fraction}_{\text{Enhanced}}}{\text{Factor of Improvement}} \right)}$$

Notation

Speedup be denoted as S

fraction_{Enhanced} be denoted as f_E

factor of improvement be denoted as f_I

Amdahl's Law can be rewritten as:

$$S = \frac{1}{(1 - f_E) + \left(\frac{f_E}{f_I} \right)}$$

$$S = ((1 - f_E) + (f_E / f_I))^{-1} .$$

Amdahl's law

Problem Type 1 – Predict System Speedup If we know f_E and f_I , then we use the Speedup equation (above) to determine S.

- **Example:** Let a program have 40 percent of its code enhanced (so $f_E = 0.4$) to run 2.3 times faster (so $f_i = 2.3$). What is the overall system speedup S?

Step 1: Setup the equation: $S = ((1 - f_E) + (f_E / f_I))^{-1}$

Step 2: Plug in values & solve $S = ((1 - 0.4) + (0.4 / 2.3))^{-1}$
 $= (0.6 + 0.174)^{-1}$
 $= 1 / 0.774$
 $= 1.292$

Amdahl's law

Problem Type 2 – Predict Speedup of Fraction Enhanced If we know f_E and S , then we solve the Speedup equation (above) to determine f_I , as follows

Example: Let a program have 40 percent of its code enhanced (so $f_E = 0.4$) to yield a system speedup 4.3 times faster (so $S = 4.3$). What is the factor of improvement f_I of the portion enhanced?

Amdahl's law

Example: Let a program have 40 percent of its code enhanced (so $f_E = 0.4$) to yield a system speedup 4.3 times faster (so $S = 4.3$). What is the factor of improvement f_I of the portion enhanced?

Case #1:

Can we do this? In other words, let's determine if by enhancing 40 percent of the system, it is possible to make the system go 4.3 times faster ...

Step 1: Assume the limit, where $f_I = \text{infinity}$, so $S = ((1 - f_E) + (f_E / f_I))^{-1} \rightarrow S = 1 / (1 - f_E)$

Step 2: Plug in values & solve $S = ((1 - 0.4))^{-1} = 1 / 0.6 = 1.67$.

Step 3: So $S = 1.67$ is the **maximum possible speedup**, and we cannot achieve $S = 4.3$!!

Amdahl's law

Case 2: Let's determine if by enhancing 40 percent of the system, it is possible to make the system go 1.3 times faster ...

Step 1: Assume the limit, where $f_I = \text{infinity}$, so $S = ((1 - f_E) + (f_E / f_I))^{-1} \rightarrow S = 1 / (1 - f_E)$

Step 2: Plug in values & solve $S = ((1 - 0.4))^{-1} = 1 / 0.6 = 1.67$.

Step 3: So $S = 1.67$ is the maximum possible speedup, and we can achieve $S = 1.3$!!

Step 4: Solve speedup equation for f_I : $1/S = (1 - f_E) + (f_E / f_I)$ [invert both sides]

$$1/S - (1 - f_E) = f_E / f_I \quad [\text{subtract } (1 - f_E)]$$

$$(1/S - (1 - f_E))^{-1} = f_I / f_E \quad [\text{invert both sides}]$$

$$f_E (1/S - (1 - f_E))^{-1} = f_I \quad [\text{multiply by } f_E]$$

Step 5: Plug in values & solve: $f_I = f_E \cdot (1/S - (1 - f_E))^{-1}$

$$= 0.4 \cdot (1/1.3 - (1 - 0.4))^{-1}$$

$$= 0.4 / (0.769 - 0.6) = 2.367$$

Step 6: Check your work: $S = ((1 - f_E) + (f_E / f_I))^{-1} = (0.6 + (0.4/2.367))^{-1} = 1.3$

Amdahl's law

Problem Type 3 – Predict Fraction of System to be Enhanced If we know f_I and S , then we solve the Speedup equation to determine f_E , as follows

- **Example:** Let a program have a portion f_E of its code enhanced to run 4 times faster (so $f_I = 4$), to yield a system speedup 3.3 times faster (so $S = 3.3$). What is the fraction enhanced (f_E)?

Step 1: This be done by assuming $f_I = \text{infinity}$, $S = 3.3 = (1 - f_E)^{-1}$ so minimum $f_E = 0.697$

Yes, this can be done for maximum f_I , so let's solve the equation to determine actual f_E

Step 2: Solve speedup equation for f_E : $S = (1 - f_E) + (f_E / f_I)^{-1}$ [state the equation]

$$3.3 = (1 - f_E) + (f_E / 4)^{-1} \quad [\text{plug in values}]$$

$$(1 - f_E) + f_E / 4 = 1/3.3 = 0.303 \quad [\text{invert both sides}]$$

$$1 - 0.75 f_E = 0.303 \quad [\text{regroup}]$$

$$0.75 f_E = 1 - 0.303 = 0.697 \quad [\text{commutativity}]$$

$$f_E = 0.697 / 0.75 = 0.929 \quad [\text{divide by 0.75}]$$

- **Step 3:** Check your work: $S = (1 - f_E) + (f_E / f_I)^{-1} = (0.071 + (0.929/4))^{-1} = 3.3$

Amdahl's law

You can also solve for F_{parallel} using Amdahl's Law if you know your speedup and the number of processors

Amdahl's law says:

$$S = \frac{T_1}{T_n} = \frac{1}{\frac{F}{n} + (1-F)} \Rightarrow \frac{1}{S} = \frac{F}{n} + (1-F) = 1 + \frac{F - nF}{n} \Rightarrow \frac{1}{S} - 1 = F \frac{(1-n)}{n}$$

Solving for F:

$$F = \frac{\frac{1}{S} - 1}{\frac{1-n}{n}} = \frac{\frac{T_n}{T_1} - 1}{\frac{1-n}{n}} = \frac{\frac{T_n - T_1}{T_1}}{\frac{1-n}{n}} = \frac{\frac{T_1 - T_n}{T_1}}{\frac{n-1}{n}} = \frac{n(T_1 - T_n)}{T_1(n-1)}$$

Use this if you know the timing

$\frac{n}{(n-1)} \frac{T_1 - T_n}{T_1}$

Use this if you know the speedup

$\frac{n}{(n-1)} \left(1 - \frac{1}{\text{Speedup}} \right)$

Amdahl's law- Examples.....

Consider 4 potential applications of the Amdahl's Law Formula:

1. 95% of a task/program/etc. is improved by 10%
2. 5% of a task/program/etc. is improved by 10X
3. 5% of a task/program/etc. is infinitely improved
4. 95% of a task/program/etc. is infinitely improved.

For all 4 cases, what is the overall speedup of the task?

$$1. \quad speedup = \frac{1}{(1-0.95) + \frac{0.95}{1.1}} = 1.094$$

$$2. \quad speedup = \frac{1}{(1-0.05) + \frac{0.05}{10}} = 1.047$$

$$3. \quad speedup = \frac{1}{(1-0.05) + \frac{0.05}{\infty}} = 1.052$$

$$4. \quad speedup = \frac{1}{(1-0.95) + \frac{0.95}{\infty}} = 20$$

Practice Problem

Question : On a large system, suppose we can upgrade a CPU to make it 50% faster for \$10,000 or upgrade its disk drives for \$7,000 to make them 150% faster. Processes spend 70% of their time running in the CPU and 30% of their time waiting for disk service. An upgrade of which component would offer the greater benefit for the lesser cost?

Practice Problem

Question 1: On a large system, suppose we can upgrade a CPU to make it 50% faster for \$10,000 or upgrade its disk drives for \$7,000 to make them 150% faster. Processes spend 70% of their time running in the CPU and 30% of their time waiting for disk service. An upgrade of which component would offer the greater benefit for the lesser cost?

$$\begin{array}{l} f = 0.70, \\ k = 1.5 \end{array} \quad S = \frac{1}{(1 - 0.7) + 0.7/1.5}$$

The **processor** option offers a **130%** speedup

$$\begin{array}{l} f = 0.30, \\ k = 2.5 \end{array} \quad S = \frac{1}{(1 - 0.3) + 0.3/2.5}$$

The **disk drive** option gives a **122%** speedup

Practice Problem

- So, upgrading the CPU would provide a 30% speedup, while upgrading the disk would provide a 22% speedup
- Recall that the CPU upgrade cost is \$10,000, and the disk upgrade cost is \$7,000
- Which is more cost effective?
- Each 1% gain in speed from upgrading the CPU costs $\frac{10000}{30} \approx \333 .
- Each 1% gain in speed from upgrading the disk costs $\frac{7000}{22} \approx \318 .
- The disk upgrade gives a smaller speedup but is more cost effective.

Thank you...